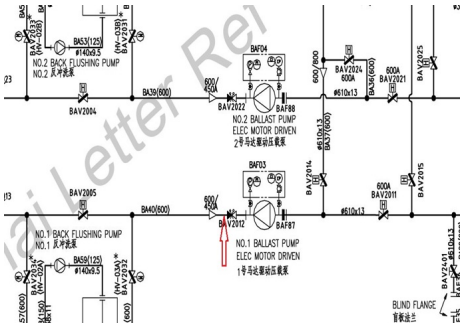
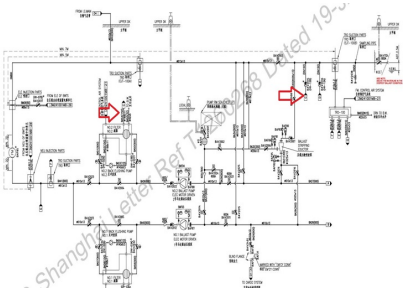


Ballast filter back flush overboard pipe

Scope:

Pin hold located at No. 2 filter back flush over board pipe needs to renew the pipe



Flange: Hole -8
PCD -240
OD - 280

TO BE INCLUDED:

- Access
- Cleaning after
- Cleaning before

MATERIALS TO BE SUPPLIED BY:

- Yard's supply

THE WORK TO BE SURVEYED BY:

- Owners representative

- Corrosion protection
- Function Testing
- Gas free certificate
- Lighting
- Paintwork
- Pressure testing
- Ventilation

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.PR.3.1	**Cost | Steel Tariff Rate**		Lumpsum	Yard
4.PR.3.4	**Cost | Installation**		Lumpsum	Yard

4.PR.3 Pipe Renewals

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide unit pipe renewal prices for the subject material and grade listed below.

1) Please provide quotation for both SCH 40 and SCH 80 – all shipyard supplied material

2) Please provide quotation for both black and hot-dip galvanized – all shipyard supplied material

- Inside tanks (e.g., cargo, ballast, fore peak, top side, wing, and aft peak etc.)
- Inside enclosed spaces (e.g., engine room, pump room etc.)
- Inside and around accommodation and accommodation decks
- Inside cargo tanks
- On main deck
- And any other area which is not covered above

Table 20: Pipe Specification

Dimension **(DN)**	1 metre of Pipe	1 pcs of Elbow	1 pcs of Bend	1 pcs of Flange	1 pcs of Sleeve	1 pcs of Bulkhead **Penetration**	1 pcs of Coupling	1 pcs of U–Bolt
1) 25								
2) 32								
3) 40								
4) 50								
5) 65								
6) 80								
7) 90								
8) 100								
9) 125								
10) 150								
11) 200								
12) 250								
13) 300								
14) 350								
15) 400								
16) 450								
17) 500								
18) 550								
19) 600								

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.PR.3.1	**Cost | Pipe**		Lumpsum	Yard
	Description: Pipe DN 250 - 18 meters			
4.PR.3.2	**Cost | Flange**		Lumpsum	Yard
4.PR.3.3	**Cost | Elbow**	2. Medium	Lumpsum	Yard
4.PR.3.4	**Cost | Pipe Installation**		Lumpsum	Yard
	Description: Provide quotation of fabrication and installation of pipe Cost to include complete fabrication as per sample with yard supplied gaskets.			

4.PR.3 Pipe Renewal | Boiler Cascade Tank

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide quotation for repairs of leaking Boiler cascade tank

Location – UPPER DECK FLOOR IN BETWEEN ACCOMODATION AND ENGINE CASING

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.PR.3.1	**Cost | Steel Tariff Rate**		Lumpsum	Yard
4.PR.3.4	**Cost | Installation**		Lumpsum	Yard

4.PR.3 Pipe Renewal | Ballast Pump Suction Strainer

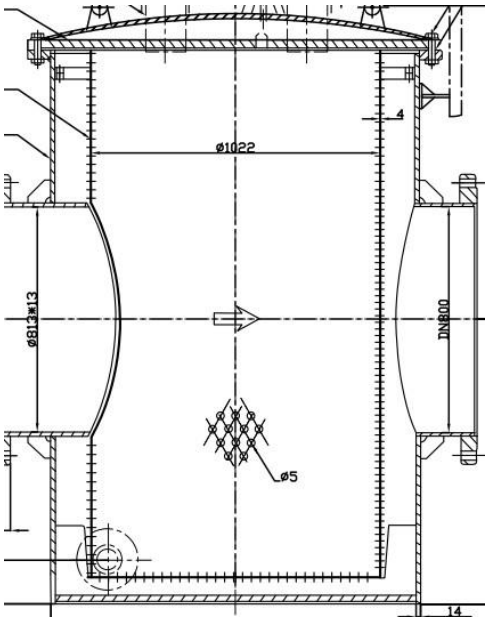
Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

REMOVE WATER BALLAST PUMPS SUCTION STRAINER AND FABRICATE NEW ONE AS PER SAMPLE TO HAVE ACCURATE DIMENSIONS.

Please provide quotation for removal of water ballast pump suction strainer and fabrication of new one

Location: UPPER DECK FLOOR IN BETWEEN ACCOMODATION AND ENGINE CASING



JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.PR.3.1	**Cost | Fabrication**		Lumpsum	Yard
4.PR.3.4	**Cost | Removal & Installation**		Lumpsum	Yard

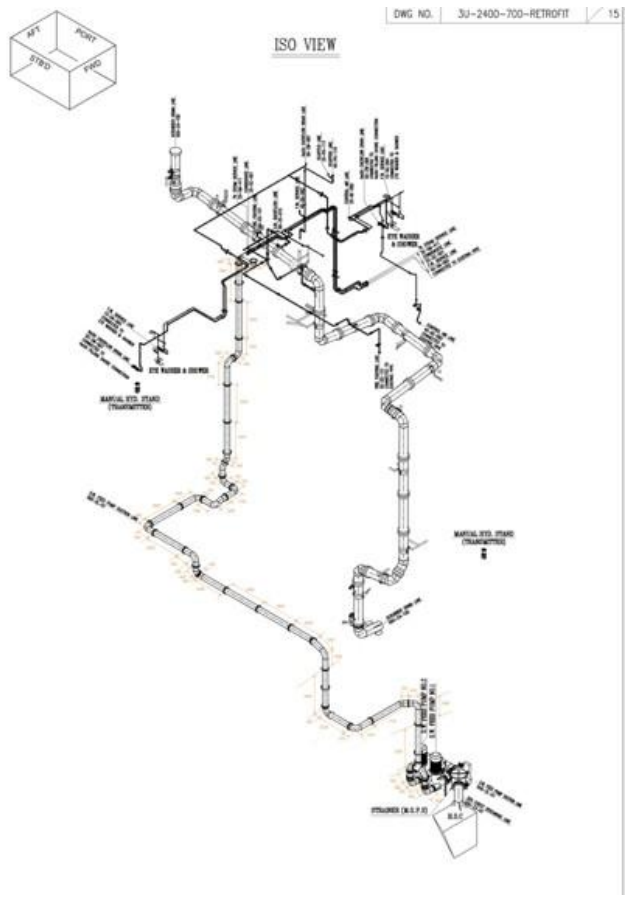
4.PR.3 Pipe Renewal | EGCS Supply Line

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide quotation for the renewal of EGCS supply line with an internal rubber coating. include pipe after auto filter.

Location: Engine room



JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.PR.3.1	**Cost | Rubber Coating**		Lumpsum	Yard
	Description: Pipe DN 250 - 18 meters			
4.PR.3.4	**Cost | Installation**		Lumpsum	Yard

4.PR.3 Pipe Renewal | EGCS Water Lance

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

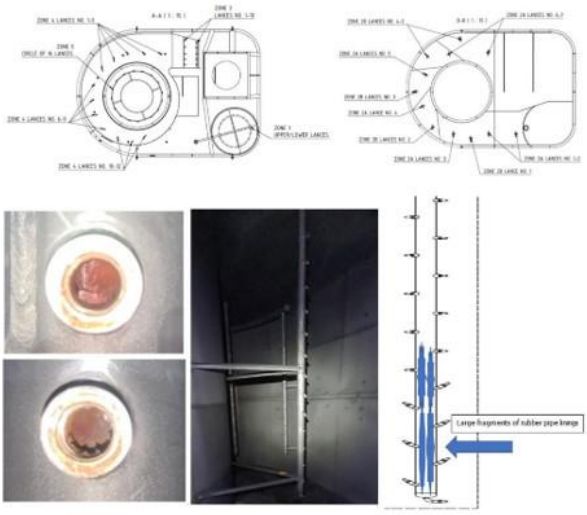
DESCRIPTION

Please provide quotation for the Installation of a water lance in the EGCS.

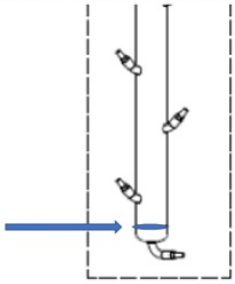
Scope:

Lower part to be cut and modified to be a flange type connection in order to access during maintenance.

Location: EGCS Tower



To cut and modify with flange type connection for maintenance



JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.PR.3.1	**Cost | Fabrication**		Lumpsum	Yard
4.PR.3.4	**Cost | Installation**		Lumpsum	Yard

4.PR.3 Pipe Renewal | Fire & GS Pump Ovb. T-Pipe

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

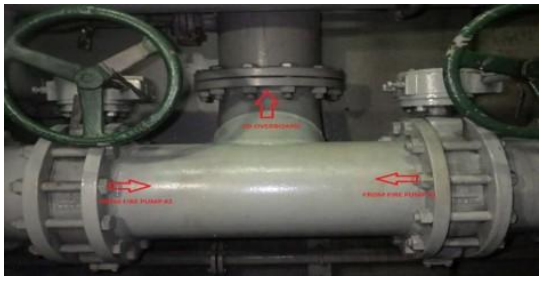
Please provide quotation for the removal and fabrication of the T-Pipe connection to Overboard

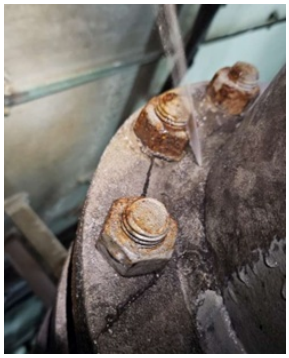
Location – ENGINE ROOM BOTTOM FLOOR

Material - GALVANIZED PIPE, SEAMLESS, SCHEDULE 80, 250A, OD-267.4 MM.

Flange - DIN 2576, 250A, PN 16, 12 HOLES x 3 pcs.

BOLTS / NUTS, NEW GASKETS.





JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.PR.3.1	**Cost | Steel Tariff Rate**		Lumpsum	Yard
4.PR.3.4	**Cost | Removal & Installation**		Lumpsum	Yard

4.VA.2 Overboard & Intermediate Valves [Storm]

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide quotation for overhauling and surveying of the following overboard and intermediate valves based on the scope described below.

1) If necessary, floor plates and pipes are to be removed for access.

2) Listed valves below are to be opened up for cleaning, inspection, overhauling and surveying.

3) Seats, discs are to be grounded, lapped, and polished. All moving parts are to be eased and greased. If necessary, butterfly valves rubber seat rings are to be renewed with owner supplied spares. Additional repairs will be decided on an individual basis by the superintendent and will be subject to additional quotation and costing.

4) During overhauling, the valves should be covered or blanked so that no blasting grit or dust from the hull maintenance program can enter the engine room.

5) While overboard valves are disconnected, shipyard is to make suitable temporary arrangements to divert any leakages from open–ended pipes to prevent filling of the machinery space bilges. Any leakages, which are not contaminated, are to be regularly pumped out using shipyard's personnel and portable equipment.

6) All sea chests and overboard valves are to be internally coated two full coats of antifouling coating.

7) Allow for all shipside butterfly valves over DN 150 are to remain in position but are to be inspected internally by using a cherry picker from the dry–dock. The cost of this inspection should include the cost of scraping the valves internally to remove any barnacles and as per the remainder of the valves coating with two coats of owner supplied antifouling coating.

8) All ship's side valves, and associated hull connections are to be surveyed in the presence of the superintendent and class surveyor.

9) All valves are to be installed back with new shipyard supplied nuts, bolts, joints, gland studs and nuts; where required, new owner supplied seat rings.

10) Where valves are removed from pipe, the same pipes are to be installed back at completion of overhauling with new shipyard supplied gaskets, bolts, and nuts.

11) If necessary, any pitted pipe sections are to be repaired by welding and grinded afterwards. These pipes are to be coated with 2 coats of owner supplied paint before installation in existing locations.

12) All the valves are to be tested in the presence of the superintendent and class surveyor. While re– floating vessel and undocking, all valves are to be tested for water tightness in the presence of the superintendent and class surveyor.

13) Any burnt areas are to be mechanically prepared to STIII and coated with 2 coats of owner supplied paint.

14) All floor plates and pipes, which were removed for access, are to be replaced back in accordance with the manufacturer's instructions and procedures.

15) The work area is to be cleaned after completion of works by shipyard without any extra charge.

Overhauling of Intermediate Valves in Workshop

Table 11: List of Overboard and Intermediate Valves

Valve Name	Location	Dimension (DN)	Type	Quantity (pcs)	Task
Valve Name	Location	Dimension (DN)	Type	Quantity (pcs)	Task
1) Sewage Treatment Disch. Valve (port)	Engine Room	1 / DN 65	Storm V/V	1	Overhaul and test in place
2) Waste Water Disch. Valve (port)	Engine Room	1 / DN 125	Storm V/V	1	Overhaul and test in place

TO BE INCLUDED: **MATERIALS TO BE SUPPLIED BY:** **THE WORK TO BE SURVEYED BY:**

- Access
- Cleaning after
- Corrosion protection
- Fire watch
- Function Testing
- Lighting
- Paintwork
- Yard's supply
- Class representative
- Owners representative

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.VA.2.3	**Cost | Overhauling DN 65**		Pieces	Yard
4.VA.2.4	**Cost | Overhauling DN 125**		Pieces	Yard

4.VA.2 Overboard & Intermediate Valves [Globe]

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide quotation for overhauling and surveying of the following overboard and intermediate valves based on the scope described below.

1) If necessary, floor plates and pipes are to be removed for access.

2) Listed valves below are to be opened up for cleaning, inspection, overhauling and surveying.

3) Seats, discs are to be grounded, lapped, and polished. All moving parts are to be eased and greased. If necessary, butterfly valves rubber seat rings are to be renewed with owner supplied spares. Additional repairs will be decided on an individual basis by the superintendent and will be subject to additional quotation and costing.

4) During overhauling, the valves should be covered or blanked so that no blasting grit or dust from the hull maintenance program can enter the engine room.

5) While overboard valves are disconnected, shipyard is to make suitable temporary arrangements to divert any leakages from open–ended pipes to prevent filling of the machinery space bilges. Any leakages, which are not contaminated, are to be regularly pumped out using shipyard's personnel and portable equipment.

6) All sea chests and overboard valves are to be internally coated two full coats of antifouling coating.

7) Allow for all shipside butterfly valves over DN 150 are to remain in position but are to be inspected internally by using a cherry picker from the dry–dock. The cost of this inspection should include the cost of scraping the valves internally to remove any barnacles and as per the remainder of the valves coating with two coats of owner supplied antifouling coating.

8) All ship's side valves, and associated hull connections are to be surveyed in the presence of the superintendent and class surveyor.

9) All valves are to be installed back with new shipyard supplied nuts, bolts, joints, gland studs and nuts; where required, new owner supplied seat rings.

10) Where valves are removed from pipe, the same pipes are to be installed back at completion of overhauling with new shipyard supplied gaskets, bolts, and nuts.

11) If necessary, any pitted pipe sections are to be repaired by welding and grinded afterwards. These pipes are to be coated with 2 coats of owner supplied paint before installation in existing locations.

12) All the valves are to be tested in the presence of the superintendent and class surveyor. While re– floating vessel and undocking, all valves are to be tested for water tightness in the presence of the superintendent and class surveyor.

13) Any burnt areas are to be mechanically prepared to STIII and coated with 2 coats of owner supplied paint.

14) All floor plates and pipes, which were removed for access, are to be replaced back in accordance with the manufacturer's instructions and procedures.

15) The work area is to be cleaned after completion of works by shipyard without any extra charge.

Overhauling of Intermediate Valves in Workshop

Table 11: List of Overboard and Intermediate Valves

Valve Name	Location	Dimension (DN)	Type	Quantity (pcs)	Task
Valve Name	Location	Dimension (DN)	Type	Quantity (pcs)	Task
1) Clean water Disch. valve (port)	Engine Room	25/DN65	Globe (Butterfly, Globe, Stop)	2	Overhaul and test in place
2) Disch. Valve from Bilge, Fire & G/S pumps line (port)	Engine Room	18 / DN 250	Globe	1	Overhaul and test in place
3) Oily W. Separator Bilge SW Disch. Valve (port)	Engine Room	25 / DN 50	Globe	2	Overhaul and test in place

Valve Name	Location	Dimension (DN)	Type	Quantity (pcs)	Task
4) Boiler Blow-OFF Disch. Valve (port)	Engine Room	25 / DN 65	Globe	2	Overhaul and test in place
5) Steam blow for Em Fire Pump Sea Chest	EFP Compartment	25 DN 25	Globe	2	Overhaul and test in place
6) Steam blow for High Suction Sea Chest	Engine Room	25 DN 25	Globe	2	Overhaul and test in place
7) Steam blow for Low Suction Sea Chest	Engine Room	25 DN 25	Globe	2	Overhaul and test in place

TO BE INCLUDED:

- Access
- Cleaning after
- Corrosion protection
- Fire watch
- Function Testing
- Lighting
- Paintwork

MATERIALS TO BE SUPPLIED BY:

- Yard's supply

THE WORK TO BE SURVEYED BY:

- Class representative
- Owners representative

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.VA.2.3	**Cost | Overhauling DN 250**		Pieces	Yard
4.VA.2.4	**Cost | Overhauling DN 50**		Pieces	Yard
	Description:			
4.VA.2.7	**Cost | Overhauling DN 65**		Pieces	Yard
4.VA.2.9	**Cost | Overhauling DN 25**		Pieces	Yard
	Description:			

4.VA.2 Overboard & Intermediate Valves [Butterfly]

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide quotation for overhauling and surveying of the following overboard and intermediate valves based on the scope described below.

1) If necessary, floor plates and pipes are to be removed for access.

2) Listed valves below are to be opened up for cleaning, inspection, overhauling and surveying.

3) Seats, discs are to be grounded, lapped, and polished. All moving parts are to be eased and greased. If necessary, butterfly valves rubber seat rings are to be renewed with owner supplied spares. Additional repairs will be decided on an individual basis by the superintendent and will be subject to additional quotation and costing.

4) During overhauling, the valves should be covered or blanked so that no blasting grit or dust from the hull maintenance program can enter the engine room.

5) While overboard valves are disconnected, shipyard is to make suitable temporary arrangements to divert any leakages from open–ended pipes to prevent filling of the machinery space bilges. Any leakages, which are not contaminated, are to be regularly pumped out using shipyard's personnel and portable equipment.

6) All sea chests and overboard valves are to be internally coated two full coats of antifouling coating.

7) Allow for all shipside butterfly valves over DN 150 are to remain in position but are to be inspected internally by using a cherry picker from the dry–dock. The cost of this inspection should include the cost of scraping the valves internally to remove any barnacles and as per the remainder of the valves coating with two coats of owner supplied antifouling coating.

8) All ship's side valves, and associated hull connections are to be surveyed in the presence of the superintendent and class surveyor.

9) All valves are to be installed back with new shipyard supplied nuts, bolts, joints, gland studs and nuts; where required, new owner supplied seat rings.

10) Where valves are removed from pipe, the same pipes are to be installed back at completion of overhauling with new shipyard supplied gaskets, bolts, and nuts.

11) If necessary, any pitted pipe sections are to be repaired by welding and grinded afterwards. These pipes are to be coated with 2 coats of owner supplied paint before installation in existing locations.

12) All the valves are to be tested in the presence of the superintendent and class surveyor. While re– floating vessel and undocking, all valves are to be tested for water tightness in the presence of the superintendent and class surveyor.

13) Any burnt areas are to be mechanically prepared to STIII and coated with 2 coats of owner supplied paint.

14) All floor plates and pipes, which were removed for access, are to be replaced back in accordance with the manufacturer's instructions and procedures.

15) The work area is to be cleaned after completion of works by shipyard without any extra charge.

Overhauling of Intermediate Valves in Workshop

Table 11: List of Overboard and Intermediate Valves

Valve Name	Location	Dimension (DN)	Type	Quantity (pcs)	Task
Valve Name	Location	Dimension (DN)	Type	Quantity (pcs)	Task
1) Clean water Disch. valve (port)	Engine Room	25/DN65	Globe (Butterfly, Globe, Stop)	2	Overhaul and test in place
2) Hi Sea Chest Suc. valve (stbd)	Engine Room	10K 800A SC	Butterfly	1	Overhaul and test in place
3) Cool. SW Disch. valve (stbd)	Engine Room	AV-HA50	Butterfly	1	Overhaul and test in place

Valve Name	Location	Dimension (DN)	Type	Quantity (pcs)	Task
4) Vacuum Cond. Disch. Valve(stbd)	Engine Room	AV-HA56	Butterfly	1	Overhaul and test in place
5) FW Generator SW Disch. Valve (port)	Engine Room	125 10K	Butterfly	1	Overhaul and test in place
6) MGO Cooler SW Disch. Valve (port)	Engine Room	DN 65 10	Butterfly	1	Overhaul and test in place
7) Low Sea Chest Suc. valve (port)	Engine Room	10K 800A SC	Butterfly	1	Overhaul and test in place
8) Dump/Drain Cooler SW Valve (port)	Engine Room	200 10K	Butterfly	1	Overhaul and test in place
9) Scrubber SW Disch. Valve (port)	Engine Room	DN 350 PN 16	Butterfly	1	Overhaul and test in place
10) AFT Draft Measuring valve (stbd)	Engine Room	10K65	Butterfly	1	Overhaul and test in place
11) Suction for Em Fire Pump (stbd)	EFP Compartment	250 10K	Butterfly	1	Overhaul and test in place
12) Ballast Overboard Valve	Pump Room	10 DN 700	Butterfly Valve Worm Gear Actuator	1	Overhaul and test in place
13) ODME Discharge Valve	Pump Room	16 DN 500	Butterfly Valve Worm Gear Actuator (Hydraulic Type)	1	Overhaul and test in place
14) Main SW Seas Chest Valve for Ballast	Pump Room	10 DN 750	Butterfly Valve Worm Gear Actuator (Hydraulic Type)	1	Overhaul and test in place
15) Cargo Seas Chest Valve	Pump Room	10 DN 600	Butterfly Valve Worm Gear Actuator (Hydraulic Type)	1	Overhaul and test in place

TO BE INCLUDED:

- Access
- Cleaning after
- Corrosion protection
- Fire watch
- Function Testing
- Lighting
- Paintwork

MATERIALS TO BE SUPPLIED BY:

- Yard's supply

THE WORK TO BE SURVEYED BY:

- Class representative
- Owners representative

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.VA.2.3	**Cost | Overhauling DN 65**		Pieces	Yard
4.VA.2.4	**Cost | Overhauling DN 125**		Pieces	Yard
4.VA.2.7	**Cost | Overhauling DN 350**		Pieces	Yard
4.VA.2.9	**Cost | Overhauling DN 700**		Pieces	Yard
	Description:			
4.VA.2.10	**Cost | Overhauling DN 500**		Pieces	Yard
	Description:			
4.VA.2.12	**Cost | Overhauling DN 750**		Pieces	Yard
	Description:			
4.VA.2.13	**Cost | Overhauling DN 600**		Pieces	Yard
	Description:			

4.VA.2.14	**Cost | Butterfly**	2. Medium	Unit	Yard
	Description: Hi Sea Chest Suc. valve (stbd) Cool. SW Disch. valve (stbd) Vacuum Cond. Disch. Valve(stbd) Low Sea Chest Suc. valve (port) Dump/Drain Cooler SW Valve (port) AFT Draft Measuring valve (stbd) Suction for Em Fire Pump (stbd)			

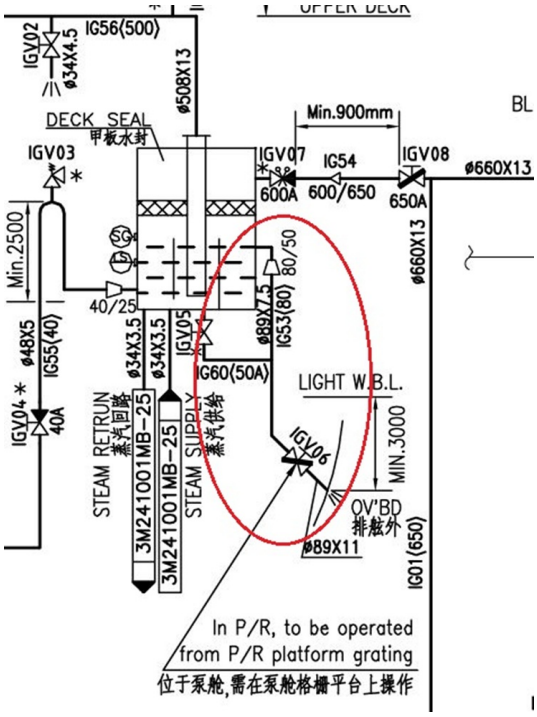
4.VA.3 Deck Seal Overboard Pipe

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	27.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Desmond Onakpojeruo	**Maker**	

DESCRIPTION

Quote for renew overboard pipes of deck seal in pump room

Deck seal overboard pipe in pumproom Size 80A Pipe lengh: 95 m Pipe material: Steel SMLS or LSAW SCH. 80 Pipe Connection: Flange, Butt weld, Flange adapter type couplings for exp. Joints Flange rate:GB -6K, SLIP -ON Gasket: Non - Asbestos Current coating: Korepex EH2352, Two component, epoxy based coating (maker: KCC Corporation, Ulsan Plan)



JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.VA.3.1	**Cost | Repairs**	2. Medium		Yard
	Description: cost for repair			

4.VA.4 Valve Actuators

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	27.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Desmond Onakpojeruo	**Maker**	

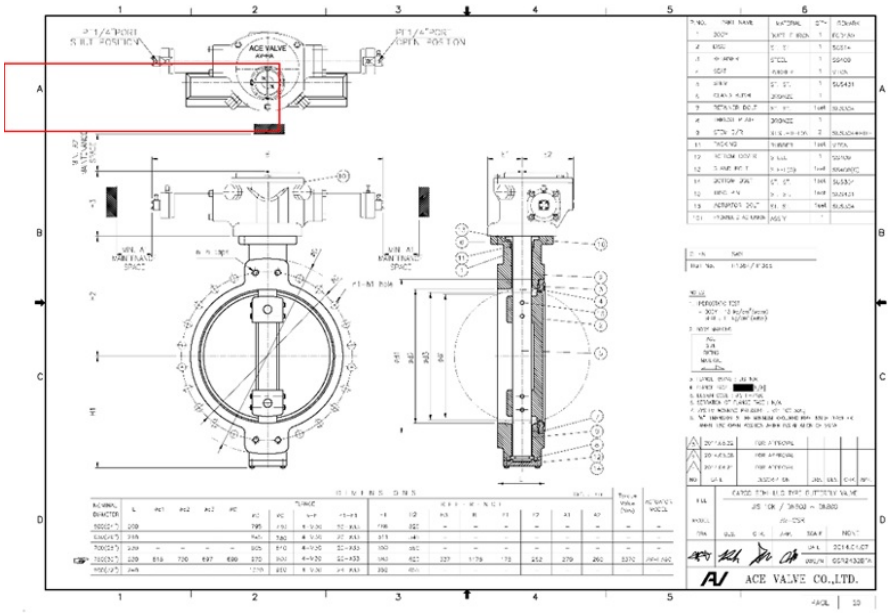
DESCRIPTION

Overhaul of valve actuator

Location – Pumproom bottom deck

Maker: ACE VALVE CO., LTD

No.	VALVE NO.	DESCRIPTION	MATERIAL				Valve size (DN)	Q'TY	CONNECTION		OPERATOR		EXTENTION		HYD. TEST (kg/cm²)		DWG' PAGE	CERT	PAINT SPEC. (IN)	PAINT SPEC. (OUT)	ACT. MODEL	CYLINDER SETTING POSITION
			BODY	DISC	SEAT	STEM			TYPE	CLASS	TYPE	POS.	H2	B	BODY	SEAT						
111	BAV2021	-	FCD450	ALBC2	NBR	SUS431	600	1EA	AV-CSR SEMI-LUG	JIS 10K	H.O	C1	-	-	15	11	31	-	Y1	U	AV-HA63	W1
112	BAV2023	-	FCD450	ALBC2	NBR	SUS431	600	1EA	AV-CSR SEMI-LUG	JIS 10K	H.O	C1	-	-	15	11	31	-	Y1	U	AV-HA63	X1
113	BAV2024	-	FCD450	ALBC2	NBR	SUS431	600	1EA	AV-CSR SEMI-LUG	JIS 10K	H.O	C2	-	-	15	11	31	-	Y1	U	AV-HA63	W1
114	BAV2025	-	FCD450	ALBC2	NBR	SUS431	600	1EA	AV-CSR SEMI-LUG	JIS 10K	H.O	C1	-	-	15	11	31	-	Y1	U	AV-HA63	X2
115	BAV2001	-	SC480	ALBC2	NBR	SUS431	800	1EA	AV-CFR FLANGE	JIS 10K	H.O	C1	-	-	15	11	30	ABS	Y1	U	AV-HA90	X2
116	BAV2002	-	FCD450	ALBC2	NBR	SUS431	800	1EA	AV-CSR SEMI-LUG	JIS 10K	H.O	C1	-	-	15	11	31	-	Y1	U	AV-HA90	X2
117	BAV2003	-	FCD450	ALBC2	NBR	SUS431	800	1EA	AV-CSR SEMI-LUG	JIS 10K	H.O	C2	-	-	15	11	31	-	Y1	U	AV-HA90	Y2
50	COV2032	-	FCD450	SCS14	VITON	SUS431	750	1EA	AV-OSR SEMI-LUG	JIS 10K	H.O	D1	-	-	15	11	20	-	K	U	AV-HA90	W1



JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
4.VA.4.1	Cost | Repairs	2. Medium		Yard

5.CL.11 Tank Bell Mouth

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Tank Bell–Mouth Inspection

The final quantity will be decided after discussion with the class surveyor.

Please provide quotation for removal of bell–mouths inside the tanks.

- The area under the bell–mouths is to be cleaned and inspected for any kind of defect.
- Thickness of inner bottom plating is to be recorded and presented to the superintendent.
- After completion of repairs, bell–mouths are to be installed back in accordance with the manufacturer's instructions and procedures with shipyard supplied gaskets, bolts, and nuts.

1)Cargo Tank Main
- Line Quantity: Unit Price - pcs
- Dimension: DN 300

2)Cargo Slop Tank Main Line
- Quantity: Unit Price - pcs
- Dimension: DN 200

3)Ballast Tank Main Line
- Quantity: Unit Price - pcs
- Dimension: DN 200

- Please confirm temporary lighting is included within the tank bell–mouth inspection quotation. Please confirm temporary ventilation is included within the tank bell–mouth inspection quotation.
- Please confirm the cost of daily gas free inspections and certificates is included within the tank bell– mouth inspection quotation.
- Please confirm the cost of final hot work certification is included within the tank bell–mouth inspection quotation.

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
5.CL.11.1	**Bell Mouth** | **Inspection**	2. Medium	Pieces	Yard

Description:
1)Cargo Tank Main Line
Quantity: Unit Price - pcs
Dimension: DN 300

Cost Structure			Total
Units	*QTY*	Unit Cost	
Per Pc		Unit Cost =	**Total Cost** [*USD*]

2)Cargo Slop Tank Main Line Quantity: Unit Price - pcs Dimension: DN 200

Cost Structure			Total
Units	*QTY*	Unit Cost	
Per Pc		Unit Cost =	**Total Cost** [*USD*]

3)Ballast Tank Main Line Quantity: Unit Price - pcs Dimension: DN 200

Cost Structure			Total
Units	*QTY*	Unit Cost	
Per Pc		Unit Cost =	**Total Cost** [*USD*]

5.CL.11.2	**Fitting** | **Repairs**	2. Medium	Pieces	Yard

Description:
For Quotation Only – Please provide quotation for filling up 30 mm x 30 mm x 10 mm size pitting by welding and grinding afterwards to remove surface defects due to new welding.
Quantity: Unit Price – pcs in Unit Price – different locations

Cost Structure			Total
Units	*QTY*	Unit Cost	
per pit		Unit Cost =	**Total Cost** [*USD*]

5.CL.11.3	**Cost** | **Cleaning & Painting**	2. Medium	m2	Yard

Description:

For Quotation Only – Please provide quotation for cleaning and coating of 1500 mm x 1500 mm section under each bell–mouth with owner supplied paint.

Quantity: Unit Price – different locations

Cost Structure			**Total**
Units	*QTY*	**Unit Cost**	
m2		Unit Cost =	**Total Cost** [*USD*]

5.CR.1 Hose Handling Crane

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

LIFTING APPLIANCES AND LOAD TESTS

Hose Handling Crane / Personnel Transfer Crane

Maker: ORIENTAL
Manufacturer: Samsung Heavy Ind., Ningbo Co.
Model: (10 T X 25 M)
Type: Electro-hydraulic driven, cylinder luffing type jib crane -HHC 420-1025
SWL: 10 ton
Test Load: 1.25 x SWL

1) Please provide quotation for arranging an authorized service engineer for crane load test in accordance with the class requirements. After being tested, all the cargo gear with the whole of the gear accessory thereto, and all chains, rings, hooks, links, shackles, swivels, pulley blocks or other loose gear are to be visually examined in place to identify whether any part has been damaged or permanently deformed by the test. If any damage or deformed condition is noted, further examination and rectification is to be affected to the satisfaction of the superintendent and class surveyor. Test certificates are to be issued and submitted to the superintendent and class surveyor.
Quantity: 1.0 – set
2) Please provide quotation for arranging an authorized service engineer for rocking test of the crane for wear down measurement of slewing bearing in accordance with the class requirements and the bearing manufacturer's instructions and procedures. Test certificates are to be issued and submitted to the superintendent and class surveyor.
Quantity: 1.0 – set
3) Please provide quotation for replacement of wires with new owner supplied spares.
Quantity: 1.0 – set
4) Please provide quotation for removal of slewing ring bolts, non–destructive testing in the presence of superintendent and class surveyor and refitting back in accordance with the manufacturer's instructions and procedures. Test certificates are to be issued and submitted to the superintendent and class surveyor.
Quantity: 1.0 – set
5) Please provide quotation for overhauling of hydraulic cylinder in shipyard workshop. Hydraulic cylinder is to be dismantled and transported to shipyard workshop. Disassemble the cylinders for inspection, overhauling, and repairs. Replace damaged and worn parts with owner supplied spares. Dust seals are to be inspected and replaced (if necessary) with owner supplied spares. After completion of repairs, perform pressure test and transport back to vessel. Refit the hydraulic cylinders in accordance with the manufacturer's instructions and procedures and perform operational test in the presence of the superintendent.
Quantity: 1.0 – pcs
6) Please provide quotation for replacement of hydraulic hoses with owner supplied spares.
Quantity: Unit Price – pcs
7) Please provide quotation for assembling staging for access and disassembling after completion of works and final inspections.
Quantity: 200 – m3 (each) in 1.0 – different locations
8) Please provide quotation for overhauling of hosting motor in shipyard workshop. After completion of repairs, perform pressure test and transport back to vessel. Refit the hosting motor in accordance with the manufacturer's instructions and procedures and perform operational test in the presence of the superintendent.
Quantity: 1.0 – pcs
9) Please provide quotation for overhauling of luffing motor in shipyard workshop. After completion of repairs, perform pressure test and transport back to vessel. Refit the luffing motor in accordance with the manufacturer's instructions and procedures and perform operational test in the presence of the superintendent. Quantity: 1.0 – pcs

DESCRIPTION	HOSE HANDLING & PERSONNEL TRANSFER CRANE	
CRANE TYPE	Hose Handling & Personnel Transfer Crane (ELEC.-HYD. DRIVEN TYPE)	
HOISTING CAPACITY	SWL 10.0 TON	SWL 1.0 TON
OVER LOAD TEST	12.5TON (RATED LOAD (SWL) × 1.25)	-
HOISTING HEIGHT	MAX 50.0 M	MAX 50.0 M
HOISTING SPEED	RATED LOAD : 10m/min (BY 2LAYER)	15m/min
	EMPTY LOAD : 20m/min (BY 2LAYER)	AT RATED LOAD (BY 2LAYER)
WORKING RADIUS	MAX. 25 M ~ MIN. 5.5 M	MAX. 26.5 M ~ MIN. 6.5 M
JIB LUFFING TIME	140 SEC. (AVERAGE UP/DOWN)	
LUFFING ANGLE	0 ° ~ 78 °	
SLEWING SPEED	MIN. 0.5 RPM	
WIRE ROPE	Ø20 MM (35x7)	Ø14 MM (35x7)
	NON-ROTATION TYPE WITH GALVANIZED SAFETY FACTOR : IN PURSUANCE FO RULE REQUIREMENT	
DESIGN CONDITION	5° HEEL + 2° TRIM	
AMBIENT TEMPERATURE	-10℃ ~ +45℃	
POWER UNIT	SELF CONTAINED TYPE MAX. WORKING PRESSURE : abt. 300 kg/㎠	
ELEC. MOTOR FOR POWER UNIT	3-PHASE SQUIRREL CAGE INDUCTION MOTOR STARTING METHOD : DIRECT ON LINE INSULATION CLASS : F ENCLOSE : IP56 (INSTALLED PEDESTAL INSIDE) MOTOR OUT PUT : 45 kW x 4P (DUTY : S6-40%, RATED 30MIN) PROTECTION : IP56 EXPLOSION PROOF TYPE (Exd IIB T4)	
ENCLOSURE	STARTER : IP23	
ELEC. SOURCE	MAIN : AC 440V, 60Hz, 3PHASE / CONTROL : AC 220V, 60Hz, 1PHASE	
TEST AND INSPECTION	DNV CERTIFICATE	
RULE AND REGULATION	DNV	
FLAG	MARSHALL ISLANDS	
SELF WEIGHT	APPROX. 29.9 TON/SET (WITHOUT HYD. OIL)	
QUANTITY	1 SET(S)/SHIP	
ALL GREASE NIPPLES	JIS B PT 1/8", PIN TYPE WITH RUBBER CAP (MAT'L : BRASS)	

TO BE INCLUDED:

- Cleaning after
- Cleaning before
- Crane
- Function Testing
- Load Test
- Magnetic Particle Inspection (MPI)
- Staging

MATERIALS TO BE SUPPLIED BY:

- Yard's supply

THE WORK TO BE SURVEYED BY:

- Class representative
- Owners representative

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
5.CR.1.1	**Rocking test**	2. Medium	Set	Yard
	Description: Please provide quotation for arranging an authorised service engineer for rocking test of the crane for wear down measurement of slewing bearing in accordance with the class requirements and the bearing manufacturer's instructions and procedures. Test certificates are to be issued and submitted to the superintendent and class surveyor. Quantity: 1.0 – set			
5.CR.1.2	**Hydraulic Cylinder O/H**	2. Medium	Pieces	Yard
	Description: Please provide quotation for overhauling of hydraulic cylinder in shipyard workshop. Hydraulic cylinder is to be dismantled and transported to shipyard workshop. Disassemble the cylinders for inspection, overhauling, and repairs. Replace damaged and worn parts with owner supplied spares. Dust seals are to be inspected and replaced (if necessary) with owner supplied spares. After completion of repairs, perform pressure test and transport back to vessel. Refit the hydraulic cylinders in accordance with the manufacturer's instructions and procedures and perform operational test in the presence of the superintendent. Quantity: 1.0 – pcs			
5.CR.1.3	**Wire Renewal**	2. Medium	Set	Yard
	Description: Please provide quotation for replacement of wires with new owner supplied spares. Quantity: 1.0 – set In case re-using of existing wire is required, 80% surcharge to be applied.			
5.CR.1.4	**Hydraulic Hoses**	2. Medium	Pieces	Yard
	Description: Please provide quotation for replacement of hydraulic hoses with owner supplied spares. Quantity: Unit Price – pcs Hydraulic cylinder hoses replacement with owner supplied material.			
5.CR.1.5	**Load Test**	2. Medium	Set	Yard
	Description: Please provide quotation for arranging an authorised service engineer for crane load test in accordance with the class requirements. After being tested, all the cargo gear with the whole of the gear accessory thereto, and all chains, rings, hooks, links, shackles, swivels, pulley blocks or other loose gear are to be visually examined in place to identify whether any part has been damaged or permanently deformed by the test. If any damage or deformed condition is noted, further examination and rectification is to be affected to the satisfaction of the superintendent and class surveyor. Test certificates are to be issued and submitted to the superintendent and class surveyor. Quantity: 1.0 – set			
5.CR.1.6	**Cost | NDT Testing**	2. Medium	Set	Yard
	Description: Please provide quotation for removal of slewing ring bolts, non–destructive testing in the presence of superintendent and class surveyor and refitting back in accordance with the manufacturer's instructions and procedures. Test certificates are to be issued in attendance of the superintendent and class surveyor. Quantity: 1.0 – set			
5.CR.1.7	**Cost | Staging**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for assembling staging for access and disassembling after completion of works and final inspections. Quantity: 200 – m3 (each) in 1.0 – different locations			
5.CR.1.8	**Hoist Motor | Overhaul**	2. Medium	Pieces	Yard
	Description: Please provide quotation for overhauling of hosting motor in shipyard workshop. After completion of repairs, perform pressure test and transport back to vessel. Refit the hosting motor in accordance with the manufacturer's instructions and procedures and perform operational test in the presence of the superintendent. Quantity: 1.0 – pcs			
5.CR.1.9	**Luffing Motor | Overhaul**	2. Medium	Set	Yard
	Description: Please provide quotation for overhauling of luffing motor in shipyard workshop. After completion of repairs, perform pressure test and transport back to vessel. Refit the luffing motor in accordance with the manufacturer's instructions and procedures and perform operational test in the presence of the superintendent.			

5.IG.6 PV Valves

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

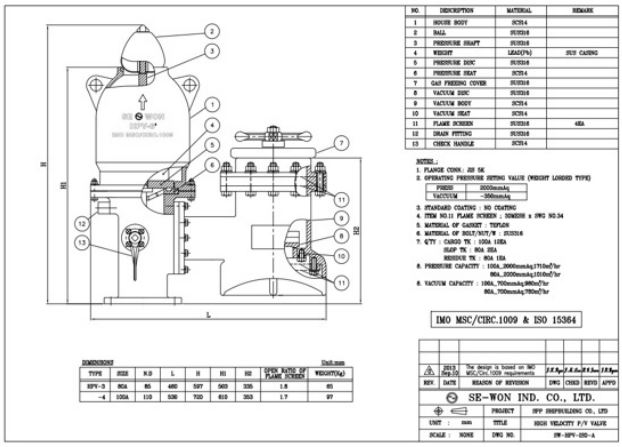
Tank PV Valves

1) Please provide quotation for overhauling of PV valves in shipyard workshop.

Quantity: 12.0 – pcs
For cargo tanks
Dimension: DN 100 Model: HPV-4

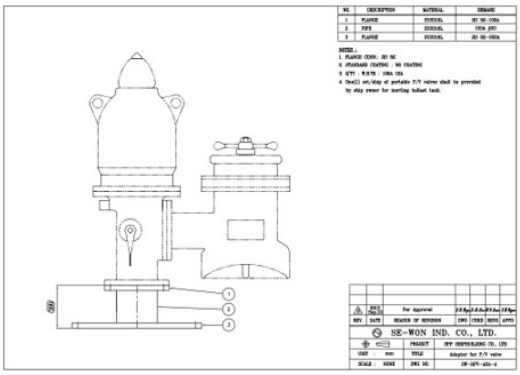
Quantity: 3.0 – pcs
Slops and Residual Dimension: DN 80
Model: HPV-3

Quantity: 2.0 – pcs
Ballast Tanks Dimension: DN 85
Model: HPV-3 1/2



2)Please provide quotation for assembling staging for access and disassembling after completion of works and final inspections.

Quantity: 16.0 – m3 (each) in 15.0 – different locations



TO BE INCLUDED:
- Cleaning after
- Function Testing
- Pressure testing
- Transport internal

MATERIALS TO BE SUPPLIED BY:
- Owner's supply
- Yard's supply

THE WORK TO BE SURVEYED BY:
- Class representative
- Owners representative

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type

5.IG.6.1	**PV Valves | Overhaul**	2. Medium	Pieces	Yard

Description:

- Please provide quotation for overhauling of PV valves in shipyard workshop. PV valves are to be disconnected from the respective piping and transferred to shipyard workshop.
- The valves are to be disassembled, cleaned and overhauled; old meshes for flame screens are to be renewed with new shipyard supplied stainless steel wire meshes.
- Pressure and vacuum test of the valves are to be performed in the presence of the superintendent and class surveyor.
- The test certificates are to be issued at the completion of the tests. After completion of repairs and tests, all the valves are to be transported back to the vessel and fitted back in accordance with the manufacturer's instructions and procedures with new shipyard supplied stainless steel bolts, nuts and Teflon gaskets.

5.IG.6.2	**PV Valves | Staging & Access**	2. Medium	m3	Yard

Description:

Please provide quotation for assembling staging for access and disassembling after completion of works and final inspections

5.IG.7 Deck Non-Return & Isolation Valve

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Deck Isolation Valve

Please provide quotation for overhauling of deck isolation valve in shipyard workshop. After completion of overhauling and repairs, a successful pressure test is to be performed in the presence of the superintendent and class surveyor. After successful pressure test, the valve is to be transported back to the vessel and installed back in accordance with the manufacturer's instructions and procedures with new shipyard supplied gaskets, stainless steel bolts and nuts.
Quantity: 1.0 – pcs Dimension: DN 300

Deck Non-Return Valve

Please provide quotation for overhauling of deck non–return valve in shipyard workshop. After completion of overhauling and repairs, a successful pressure test is to be performed in the presence of the superintendent and class surveyor. After successful pressure test, the valve is to be transported back to the vessel and installed back in accordance with the manufacturer's instructions and procedures with new shipyard supplied gaskets, stainless steel bolts and nuts.
Quantity: 1.0 – pcs Dimension: DN 300

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
5.IG.7.1	**Cost | overhaul**	2. Medium	Pieces	Yard

Description:

- Please provide quotation for overhauling of valves in shipyard workshop.
- After completion of overhauling and repairs, a successful pressure test is to be performed in the presence of the superintendent and class surveyor.
- After successful pressure test, the valve is to be transported back to the vessel and installed back in accordance with the manufacturer's instructions and procedures with new shipyard supplied gaskets, stainless steel bolts and nuts.

5.PM.1 COPT Overhaul

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	27.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Desmond Onakpojeruo	**Maker**	

DESCRIPTION

COPT Overhaul Specification

General Tasks (All COPT Units 1, 2 & 3):

Overhaul all three COPT units.

Inspect condition of nozzle rings.

Renew all fastening bolts for nozzle rings.

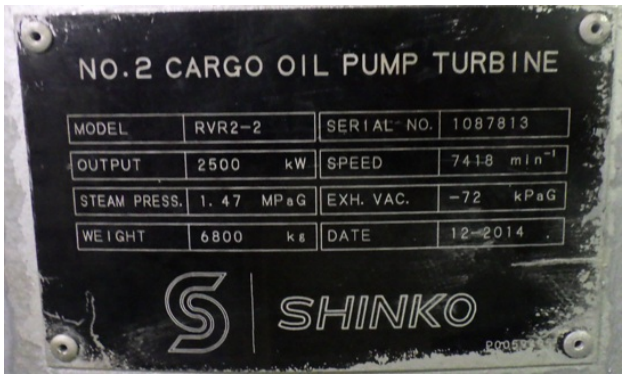
COPT Job scope.

1- COP 1 and 2 rotor replacement

2- COP1,2, and 3 nozzle ring reconditioning.

3- All three COPs will be sifted to shipyard for full reconditioning. For this, shipside will need to crop to make window to take the COP out.

Location – ENGINE ROOM. COPT 3rd deck FWD.



JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
5.PM.1.1	**Cost | Overhaul**	2. Medium	Lumpsum	Yard
	Description: Quote cost for overhaul			

5.TC.1 Foam Monitors

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

1) Please provide quotation for disassembly and replace foam monitors bearings with owner supply, foam monitors are to be transferred back to the vessel and delivered to the superintendent. Full-service report is to be submitted to superintendent. 2) Please provide quotation for machining IWO of flanges (assume max diameter 200~250 mm) in case needed in a boring machine. 3) Complete overhauling of all monitors (6 pcs) required with polishing and adjustment 4) Replacement of all O-Rings, Bolts, Nuts, grease nipples

TO BE INCLUDED:
- Function Testing

MATERIALS TO BE SUPPLIED BY:
- Owner's supply

THE WORK TO BE SURVEYED BY:
- Class representative
- Owners representative

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
5.TC.1.1	**Form Monitors Overhaul**	2. Medium	Pieces	Yard

5.TC.1 Tank Cleaning Machine

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no	PJ 30	**Component**	TANK CLEANINIG MACHINE
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	POLARMARINE

DESCRIPTION

Tank Cleaning Machine

Manufacturer: POLARMARINE Model: PJ 30
Type: DECK MOUNTING SINGLE NOZZLE TURBIND DRIVEN UNIT
Capacity: 30.0 m3 per hour

Please provide quotation for overhauling of tank cleaning machine in shipyard workshop.

TO BE INCLUDED:
- Function Testing

MATERIALS TO BE SUPPLIED BY:
- Owner's supply

THE WORK TO BE SURVEYED BY:
- Class representative
- Owners representative

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
5.TC.1.1	**Tank Cleaning Machine Overhaul**		Unit	Yard

Description:

- All the connections and switches are to be disconnected from the tank cleaning machine and blanked off to prevent any leakage around the tank, main deck, and tank cleaning machine.
- All clearances are to be taken, recorded, and submitted to the superintendent prior to and after completion of overhauling.
- In the shipyard workshop, all the parts such as control unit, turbine unit, washing unit are to be dismantled and opened for inspection and overhauling.
- All parts are to be cleaned completely for inspection and survey.
- All worn out parts such as bushes, mechanical seals, turbine, etc. are to be replaced with owner supplied spares.
- After completion of overhauling and repairs, the tank cleaning machine is to be assembled and pressure tested in the presence of superintendent and class surveyor.
- After successful pressure test, the pump is to be transported back to the vessel and installed back in accordance with the manufacturer's instructions and procedures with new shipyard supplied gaskets, bolts, and nuts.
- After completion of installation, a successful operational test is to be performed in the presence of the superintendent and class surveyor.
- Full-service report is to be submitted to superintendent.

Quantity: 14.0 – pcs

6.BT.1 Steel Renewal | in Hull Internals

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Preamble for Structural Steelwork Repair Items

The following are to be allowed for using your Standard Tariff prices for “all in” rates per kilo:
1) Shaping and corrugation of shell plates etc.
2) Access work
3) Ventilation
4)Lighting
5) Staging
6) Protection of adjacent plates
7) Fairing of adjacent plates
8) Grit blasting to SA 2.5 ISO 8501-1, preparation, and coating as original of new and disturbed steel and equipment (Owner supplied paint)
9) Classification and Owner's NDT requirements
10) Gas Free Certificates and chemist’s visits
11) Fire watch in way of repair area
12) All pressure tests and retests
13) Removal and disposal of debris, cleaning after repairs
14) X–rays
15) Rigging, crane service, compressed air supply
16) Delivery of all material and equipment to and from the vessel
17) Steel specific gravity is to be taken as 7.85 or as otherwise specified.

General Notes

General notes with reference to steel renewal work:

1) Where defective steelwork is cut away, the resultant edges are to be ground smooth and edge prepared where required.
2) Where plating only is removed from bulkheads, the remaining bulkhead stiffener edges are to be ground smooth prior to refitting the plating. Where stiffeners only are removed from the bulkheads, the plating in way of the removed stiffeners is to be ground smooth prior to fitting the new stiffener.
3) New steel is to be fitted and properly faired with due care to obtain proper alignment. Superintendent is to witness “fit up” condition.
4) Any high tensile steel is to be pre–heated immediately prior to welding as per classification requirements. Approved low hydrogen electrodes are to be used.
5) Superintendent is to see back gouging prior to final welding.
6) All butt–welding is to be full penetration welding with proper edge preparation.
7) Electrodes approved by the vessel’s Classification Society are to be used throughout the repair.
8) Your proposed welding sequence is to be submitted for approval to owner and Class. All welding practice and sequences are to be carried out with due care hence minimizing welding stresses. Care has also to be taken to ensure all welding is returned in way of plate edges at slots, scallops etc.
9) All steel fabrication and renewals are to be carefully prepared prior to painting. All weld spatters are to be removed, all sharp edges rounded, weld bead dressed, and scars removed, etc.
10) Where horizontal stringers are repaired by fitting inserts, the top weld reinforcement is to be ground flush. Welding is to be dye penetrant tested prior to any coating. Where inserts are fitted in way of original access and drainage holes, these holes are to be reinstated.

11) Edges of all holes, slots, etc. to be ground smooth with no serration. Particular care is to be taken at scallops.
12) Fabrication steel sections are to be grit blasted to SA 2.5 ISO 8501-1, inspected, and then coated ashore prior to delivery on board. Final coating is also to be inspected. All steel used for deck and shell repairs are to be supplied on–board, blasted and primed after fitting having 2 primers and 1 finish coats.
13) All steel work is to be repaired or renewed as stated in this specification subject to on site examination by the Superintendent before authorisation. The dimensions indicated can be used for your estimating purposes.
14) Prior to any work being carried out, repairs are to be “lined off” on–board and approved by Classification and the Superintendent.

15) Some repairs are for repair and renewal of plate work only, not plate plus internals. The “all in” kilo rates should also include “plate work only” repairs rates. Some repairs may consist of repair and renewal of internals only, not plate plus internals. Your “all in” kilo rates should also include “internals only” repairs rate. You are to confirm in your quotation that all steel material, staging, and other equipment required to carry out the repairs as indicated is available. You should also indicate if further material is available and to take account of a reasonable amount of additional work which may be found.

16) High tensile steel may in some cases be required and all steel used must have Certificates from the
vessel’s Classification Society.
17) There may be an isolated repair, which will be repaired in a small section. You are to confirm that these small piece repairs will be charged at the “all in” kilo rate. If you require a different kilo rate, this rate per kilo is to be quoted.

18) If a different rate is required you should confirm that where the repair is accessible from “all in” kilo rate staging, that no additional charge for staging is included.
19) All staging for steel repairs will be included in “all in” rate per kilo for steel. The repair facility is to make a tank–by–tank drawing of any additional staging for access or cleaning or any other purpose when this staging is erected. This drawing is to show the location of the staging, with dimension on which the cost will be based.
20) The details on this drawing are to be agreed by the repair facility and Superintendent prior to the staging being dismantled.

21) Classification surveyors and Superintendent are to sight and obtain copies of all plate certificates and sight plate stamping prior to cutting for the vessel. If certificates are presented after the plate is cut, they may not be accepted.
22) Any piping used in this specification for handrails or guardrails is to be heavy quality, minimum schedule 40.

Scope of Steel Repair Works

1) Inside tanks (e.g., cargo, ballast, fore peak, top side, wing, and aft peak etc.)
2) Inside enclosed spaces (e.g., engine room, pump room etc.)
3) Inside and around accommodation and accommodation decks
4) Inside cargo holds (if applicable)
5) On main deck
6) An any other area which is not covered above

PLEASE CONSIDER/QUOTE FOR 35 TONS OF STEEL WORK AH32 INSIDE BALLAST TANKS DISTRIBUTED TO 14 DIFFERENT BALLAST TANKS AND MOSTLY 25KG OR BELOW

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
6.BT.1.1	**Cost | Steel Tariff Rate**	2. Medium	Kilograms	Yard
	Description: Steel Grades and Locations Please provide unit steel prices for the subject material and grade listed in cargo tanks, ballast tanks (topside, wing, and double bottom), aft peak, forepeak, accommodation, main deck, and any other location that the shipyard has increment in the unit price. 1) A grade straight plates, single curved, double curved, and profiles 2) B grade straight plates, single curved, double curved, and profiles 3) AH32 grade straight plates, single curved, double curved, and profiles 4) AH36 grade straight plates, single curved, double curved, and profiles ***all applicable surcharges to be specified.*			
6.BT.1.2	**Cost | FW Washing**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for fresh water washing and mechanical surface preparation to ST 3.0 standard of new steel plates and burnt areas and apply 2 coats of owner supplied paint (both sides of the plates).			
6.BT.1.3	**Cost | NDT Welding**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for NDT of welding seams			
6.BT.1.4	**Cost | Vaccum Test**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for vacuum test of welding seams			
6.BT.1.5	**Cost | X-Ray**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for X–Ray of welding seams			
6.BT.1.6	**Cost | Tank air testing**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for tank air testing to 0.2 bars			

6.BT.1 Steel Renewal | in Ballast Tanks

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide service fees and units cost (kg) for the steel renewal works including all applicable surcharges.

Below mentioned general notes provide detailed instructions and considerations for steel renewal work during drydock.

General Notes

Edge Preparation and Grinding:
Defective steelwork edges are to be ground smooth and edge-prepared as required.

Bulkhead and Stiffener Renewal
Renewal of steel involves proper fairing, alignment, and witness "fit up" conditions.

Welding Practices
High tensile steel is to be pre-heated as per classification requirements.
Back gouging is to be seen by the Superintendent before final welding.
Full penetration welding with proper edge preparation is required.
Approved low hydrogen electrodes and welding sequences are to be used and submitted for approval.

Post-Weld Treatment
Steel fabrication and renewals are to be prepared before painting, including the removal of weld spatters, rounding sharp edges, dressing weld beads, and removing scars.

Hole and Slot Preparation
Edges of all holes, slots, etc., are to be ground smooth with no serration.

Grit Blasting and Coating
Fabricated steel sections are to be grit blasted to SA 2.5 ISO 8501-1, inspected, and coated ashore before delivery.
All steel used for deck and shell repairs is to be supplied on-board, blasted, and primed after fitting.

Scope

Please provide quotation for removal of damaged U-Bolts, material supply, and installation of new U- Bolts as per original dimensions and material. After completion of installation, the surface of the damaged and burnt areas are to be mechanically prepared to ST 3.0 standard and coated with owner supplied paint.

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
6.BT.1.1	**Cost | Steel Tariff Rate**	2. Medium	Kilograms	Yard
	Description: ***all applicable surcharges to be specified.*			
6.BT.1.2	**Vertical Ladders**	2. Medium	Unit	Yard
	Description: Ballast tanksvertical ladders Renewal (for quotation only) not applicable 1) Please provide quotation for removal of damaged vertical ladders, material supply, manufacture in shipyard workshop, and installation of new vertical ladders as per original dimensions and material. After completion of manufacturing, the vertical ladders are to be grit blasted to SA 2.0 ISO 8501-1 and coated with owner supplied paint. After completion of installation, the surface of the burnt areas are to be mechanically prepared to ST 3.0 standard and coated with owner supplied paint. Quantity: TBC – sets Dimensions: TBC			
6.BT.1.3	**U-Bolts**	2. Medium	Unit	Yard
	Description: Please provide quotation for removal of damaged U-Bolts, material supply, and installation of new U- Bolts as per original dimensions and material. After completion of installation, the surface of the damaged and burnt areas are to be mechanically prepared to ST 3.0 standard and coated with owner supplied paint. U-Bolts Size: 350A, 300A, 250A, 200A, 150A & 65A. Share quotes as per size per piece renewal including Material yard supply. Quantity: TBC – sets Location: TBC Material: TBC			

6.BT.1 Steel Renewal | in Hull

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Preamble for Structural Steelwork Repair Items

The following are to be allowed for using your Standard Tariff prices for “all in” rates per kilo:
1) Shaping and corrugation of shell plates etc.
2) Access work
3) Ventilation
4)Lighting
5) Staging
6) Protection of adjacent plates
7) Fairing of adjacent plates
8) Grit blasting to SA 2.5 ISO 8501-1, preparation, and coating as original of new and disturbed steel and equipment (Owner supplied paint)
9) Classification and Owner's NDT requirements
10) Gas Free Certificates and chemist’s visits
11) Fire watch in way of repair area
12) All pressure tests and retests
13) Removal and disposal of debris, cleaning after repairs
14) X–rays
15) Rigging, crane service, compressed air supply
16) Delivery of all material and equipment to and from the vessel
17) Steel specific gravity is to be taken as 7.85 or as otherwise specified.

General Notes

General notes with reference to steel renewal work:

1) Where defective steelwork is cut away, the resultant edges are to be ground smooth and edge prepared where required.
2) Where plating only is removed from bulkheads, the remaining bulkhead stiffener edges are to be ground smooth prior to refitting the plating. Where stiffeners only are removed from the bulkheads, the plating in way of the removed stiffeners is to be ground smooth prior to fitting the new stiffener.
3) New steel is to be fitted and properly faired with due care to obtain proper alignment. Superintendent is to witness “fit up” condition.
4) Any high tensile steel is to be pre–heated immediately prior to welding as per classification requirements. Approved low hydrogen electrodes are to be used.
5) Superintendent is to see back gouging prior to final welding.
6) All butt–welding is to be full penetration welding with proper edge preparation.
7) Electrodes approved by the vessel’s Classification Society are to be used throughout the repair.
8) Your proposed welding sequence is to be submitted for approval to owner and Class. All welding practice and sequences are to be carried out with due care hence minimizing welding stresses. Care has also to be taken to ensure all welding is returned in way of plate edges at slots, scallops etc.
9) All steel fabrication and renewals are to be carefully prepared prior to painting. All weld spatters are to be removed, all sharp edges rounded, weld bead dressed, and scars removed, etc.
10) Where horizontal stringers are repaired by fitting inserts, the top weld reinforcement is to be ground flush. Welding is to be dye penetrant tested prior to any coating. Where inserts are fitted in way of original access and drainage holes, these holes are to be reinstated.

11) Edges of all holes, slots, etc. to be ground smooth with no serration. Particular care is to be taken at scallops.
12) Fabrication steel sections are to be grit blasted to SA 2.5 ISO 8501-1, inspected, and then coated ashore prior to delivery on board. Final coating is also to be inspected. All steel used for deck and shell repairs are to be supplied on–board, blasted and primed after fitting having 2 primers and 1 finish coats.
13) All steel work is to be repaired or renewed as stated in this specification subject to on site examination by the Superintendent before authorisation. The dimensions indicated can be used for your estimating purposes.
14) Prior to any work being carried out, repairs are to be “lined off” on–board and approved by Classification and the Superintendent.

15) Some repairs are for repair and renewal of plate work only, not plate plus internals. The “all in” kilo rates should also include “plate work only” repairs rates. Some repairs may consist of repair and renewal of internals only, not plate plus internals. Your “all in” kilo rates should also include “internals only” repairs rate. You are to confirm in your quotation that all steel material, staging, and other equipment required to carry out the repairs as indicated is available. You should also indicate if further material is available and to take account of a reasonable amount of additional work which may be found.

16) High tensile steel may in some cases be required and all steel used must have Certificates from the
vessel’s Classification Society.
17) There may be an isolated repair, which will be repaired in a small section. You are to confirm that these small piece repairs will be charged at the “all in” kilo rate. If you require a different kilo rate, this rate per kilo is to be quoted.

18) If a different rate is required you should confirm that where the repair is accessible from “all in” kilo rate staging, that no additional charge for staging is included.
19) All staging for steel repairs will be included in “all in” rate per kilo for steel. The repair facility is to make a tank–by–tank drawing of any additional staging for access or cleaning or any other purpose when this staging is erected. This drawing is to show the location of the staging, with dimension on which the cost will be based.
20) The details on this drawing are to be agreed by the repair facility and Superintendent prior to the staging being dismantled.

21) Classification surveyors and Superintendent are to sight and obtain copies of all plate certificates and sight plate stamping prior to cutting for the vessel. If certificates are presented after the plate is cut, they may not be accepted.
22) Any piping used in this specification for handrails or guardrails is to be heavy quality, minimum schedule 40.

Scope of Steel Repair Works on Hull

The welding seams suffer from wastage of the welds cap and undercut in way of the fusion line. Affected weld runs are to be back gouged to clean metal, then filled and capped with weld runs back to the original profile. Repaired welds are to be inspected by superintendent and class surveyor. The welding procedure and all welding materials also must be class approved.
The final quantity is to be decided after inspection in dry–dock

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
6.BT.1.1	**Cost | Steel Tariff Rate**	2. Medium	Kilograms	Yard
	Description: Quantity: 200.0 meter – assuming 50.0 meter x 4.0 welding runs (total is 50.0 x 4.0 welding runs = 200.0 m) ***all applicable surcharges to be specified.*			
6.BT.1.2	**Cost | Back gouging**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for back–gouging, cleaning, and welding of wasted welding seams. Quantity: 200.0 meter – assuming 50.0 meter x 4.0 welding runs (total is 50.0 x 4.0 welding runs = 200.0 m) Please confirm staging, crane, and cherry picker assistance is included within the back–gouging, cleaning, and welding of wasted welding seams quotation.			
6.BT.1.3	**Cost | NDT Welding**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for NDT of welding seams			
6.BT.1.4	**Cost | Vaccum Test**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for vacuum test of welding seams			
6.BT.1.5	**Ballast Tank 6S Dent**	2. Medium	Lumpsum	Yard
	Description: THE VERTICAL DENT OF SIDE SHELL IN PLACE BETWEEN L21-11 AND L21-14. DIMENTION 1500 MM LONG AND 1270 MM HIGH, THE SHALL IS CONCAVED INTO TANK ON DEPTH AROUND 45 MM AND STRUCTURALY DEFORMED THE LONGITUDINAL STIFF.			
6.BT.1.6	**Cost | Grit Blasting**	2. Medium	Lumpsum	Yard
	Description: For Quotation Only – Please provide quotation for grit blasting of welding seams to SA 2.0 ISO 8501- 1			
6.BT.1.7	**Ballast Tank 2S Dent**	2. Medium	Lumpsum	Yard
	Description: Vertical frame dent on the 2nd platform near ladder between stringers, length 50 cm, deep abt 3.5 cm. Top deck dent in fwd part 1st platform, length 1.30mtr, deep abt 2cm			
6.BT.1.8	**Main Deck 2S Ballast Tank Area**	2. Medium	Lumpsum	Yard
	Description: Length 1 meter , width 20 cm , deep 3 cm			
6.BT.1.9	**Ballast Tank 3P Dent**	2. Medium	Lumpsum	Yard
	Description: Stringer's dent on the 2nd platform very FWD part, length 40 cm, deep abt 2.5 cm.			

6.SF.1 Steel Renewal | VRCs Brackets & Covers

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide service fees and units cost (kg) for the steel renewal works including all applicable surcharges.

Below mentioned general notes provide detailed instructions and considerations for steel renewal work during drydock.

General Notes

Edge Preparation and Grinding:
Defective steelwork edges are to be ground smooth and edge-prepared as required.

Bulkhead and Stiffener Renewal
Renewal of steel involves proper fairing, alignment, and witness "fit up" conditions.

Welding Practices
High tensile steel is to be pre-heated as per classification requirements.
Back gouging is to be seen by the Superintendent before final welding.
Full penetration welding with proper edge preparation is required.
Approved low hydrogen electrodes and welding sequences are to be used and submitted for approval.

Post-Weld Treatment
Steel fabrication and renewals are to be prepared before painting, including the removal of weld spatters, rounding sharp edges, dressing weld beads, and removing scars.

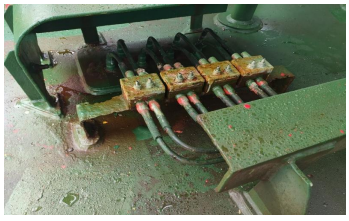
Hole and Slot Preparation
Edges of all holes, slots, etc., are to be ground smooth with no serration.

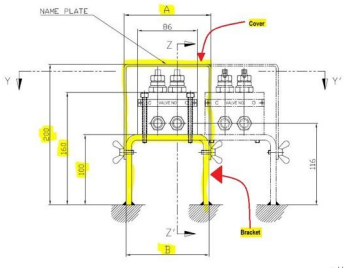
Grit Blasting and Coating
Fabricated steel sections are to be grit blasted to SA 2.5 ISO 8501-1, inspected, and coated ashore before delivery.
All steel used for deck and shell repairs is to be supplied on-board, blasted, and primed after fitting.

Scope

Please provide quotation for the renewal of all valve remote control system quick connection blocks bracket and covers

Location: Various locations on main deck





BLOCK Q'TY / DIMENSION	1	2	3	4	5	6
A	126	223	320	417	514	611
B	120	217	314	411	508	605
BRACKET & COVER Q'TY	2	13	–	–	–	6

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
6.SF.1.1	**Cost | Steel Tariff Rate**	2. Medium	Kilograms	Yard
	Description: ***all applicable surcharges to be specified.*			
6.SF.1.2	**Cost | Installation**	2. Medium	Lumpsum	Yard
	Description: TO WELD A DIA 20MM X 1500MM ROUND BAR TO BOTH STANCHIONS			

6.SF.1 Steel Renewal | IGS Scrubber

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide service fees and units cost (kg) for the steel renewal works including all applicable surcharges.

Below mentioned general notes provide detailed instructions and considerations for steel renewal work during drydock.

General Notes

Edge Preparation and Grinding:
Defective steelwork edges are to be ground smooth and edge-prepared as required.

Bulkhead and Stiffener Renewal
Renewal of steel involves proper fairing, alignment, and witness "fit up" conditions.

Welding Practices
High tensile steel is to be pre-heated as per classification requirements.
Back gouging is to be seen by the Superintendent before final welding.
Full penetration welding with proper edge preparation is required.
Approved low hydrogen electrodes and welding sequences are to be used and submitted for approval.

Post-Weld Treatment
Steel fabrication and renewals are to be prepared before painting, including the removal of weld spatters, rounding sharp edges, dressing weld beads, and removing scars.

Hole and Slot Preparation
Edges of all holes, slots, etc., are to be ground smooth with no serration.

Grit Blasting and Coating
Fabricated steel sections are to be grit blasted to SA 2.5 ISO 8501-1, inspected, and coated ashore before delivery.
All steel used for deck and shell repairs is to be supplied on-board, blasted, and primed after fitting.

Scope

Please provide quotation for the repair of all internal and external damage on the IGS Scrubber Tower

Location: Engine room Funnel Casing (Upper deck)

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
6.SF.1.1	**Cost | Steel Tariff Rate**	2. Medium	Kilograms	Yard
	Description: ***all applicable surcharges to be specified.*			
6.SF.1.2	**Cost | Installation**	2. Medium	Lumpsum	Yard
	Description: TO WELD A DIA 20MM X 1500MM ROUND BAR TO BOTH STANCHIONS			

6.SF.1 Steel Renewal | PV Breaker Swivel

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide service fees and units cost (kg) for the steel renewal works including all applicable surcharges.

Below mentioned general notes provide detailed instructions and considerations for steel renewal work during drydock.

General Notes

Edge Preparation and Grinding:
Defective steelwork edges are to be ground smooth and edge-prepared as required.

Bulkhead and Stiffener Renewal
Renewal of steel involves proper fairing, alignment, and witness "fit up" conditions.

Welding Practices
High tensile steel is to be pre-heated as per classification requirements.
Back gouging is to be seen by the Superintendent before final welding.
Full penetration welding with proper edge preparation is required.
Approved low hydrogen electrodes and welding sequences are to be used and submitted for approval.

Post-Weld Treatment
Steel fabrication and renewals are to be prepared before painting, including the removal of weld spatters, rounding sharp edges, dressing weld beads, and removing scars.

Hole and Slot Preparation
Edges of all holes, slots, etc., are to be ground smooth with no serration.

Grit Blasting and Coating
Fabricated steel sections are to be grit blasted to SA 2.5 ISO 8501-1, inspected, and coated ashore before delivery.
All steel used for deck and shell repairs is to be supplied on-board, blasted, and primed after fitting.

Scope

Please provide quotation for the renewal of the bracket of the fixed swivel davit of the PV Breaker

Location: Mid ship next to PV Breaker

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
6.SF.1.1	**Cost | Steel Tariff Rate**	2. Medium	Kilograms	Yard
	Description: ***all applicable surcharges to be specified.*			
6.SF.1.2	**Cost | Installation**	2. Medium	Lumpsum	Yard
	Description: TO WELD A DIA 20MM X 1500MM ROUND BAR TO BOTH STANCHIONS			

6.SF.1 Steel Renewal | BWTS Drain Sys.

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide service fees and units cost (kg) for the steel renewal works including all applicable surcharges.

Below mentioned general notes provide detailed instructions and considerations for steel renewal work during drydock.

General Notes

Edge Preparation and Grinding:
Defective steelwork edges are to be ground smooth and edge-prepared as required.

Bulkhead and Stiffener Renewal
Renewal of steel involves proper fairing, alignment, and witness "fit up" conditions.

Welding Practices
High tensile steel is to be pre-heated as per classification requirements.
Back gouging is to be seen by the Superintendent before final welding.
Full penetration welding with proper edge preparation is required.
Approved low hydrogen electrodes and welding sequences are to be used and submitted for approval.

Post-Weld Treatment
Steel fabrication and renewals are to be prepared before painting, including the removal of weld spatters, rounding sharp edges, dressing weld beads, and removing scars.

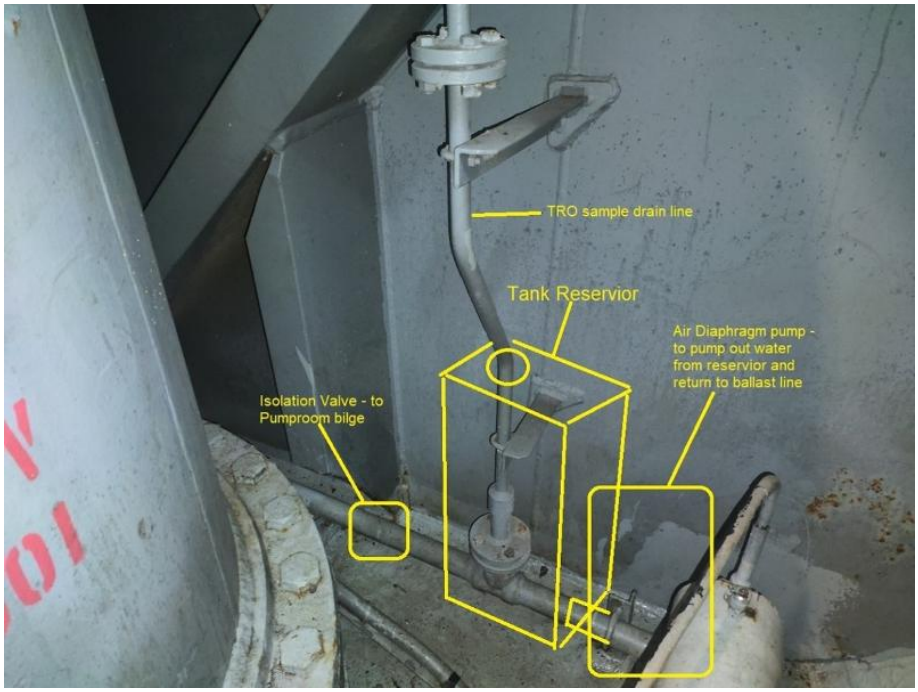
Hole and Slot Preparation
Edges of all holes, slots, etc., are to be ground smooth with no serration.

Grit Blasting and Coating
Fabricated steel sections are to be grit blasted to SA 2.5 ISO 8501-1, inspected, and coated ashore before delivery.
All steel used for deck and shell repairs is to be supplied on-board, blasted, and primed after fitting.

Scope

Please provide quotation for the Fabrication and installation of a 30L Tank reservoir and Pneumatic Diaphragm Pump for BWTS tro drain

Location: Pump room Bottom



JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
6.SF.1.1	**Cost | Steel Tariff Rate**	2. Medium	Kilograms	Yard
	Description: ***all applicable surcharges to be specified.*			
6.SF.1.2	**Cost | Installation**	2. Medium	Lumpsum	Yard
	Description: TO WELD A DIA 20MM X 1500MM ROUND BAR TO BOTH STANCHIONS			
6.SF.1.3	**Cost | Fabrication**	2. Medium	Lumpsum	Yard

6.SF.1 Steel Renewal | Slop P & S Steam Pipes

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide service fees and units cost (kg) for the steel renewal works including all applicable surcharges.

Below mentioned general notes provide detailed instructions and considerations for steel renewal work during drydock.

General Notes

Edge Preparation and Grinding:
Defective steelwork edges are to be ground smooth and edge-prepared as required.

Bulkhead and Stiffener Renewal
Renewal of steel involves proper fairing, alignment, and witness "fit up" conditions.

Welding Practices
High tensile steel is to be pre-heated as per classification requirements.
Back gouging is to be seen by the Superintendent before final welding.
Full penetration welding with proper edge preparation is required.
Approved low hydrogen electrodes and welding sequences are to be used and submitted for approval.

Post-Weld Treatment
Steel fabrication and renewals are to be prepared before painting, including the removal of weld spatters, rounding sharp edges, dressing weld beads, and removing scars.

Hole and Slot Preparation
Edges of all holes, slots, etc., are to be ground smooth with no serration.

Grit Blasting and Coating
Fabricated steel sections are to be grit blasted to SA 2.5 ISO 8501-1, inspected, and coated ashore before delivery.
All steel used for deck and shell repairs is to be supplied on-board, blasted, and primed after fitting.

Scope

Renew the steam pipe on the slop Port and STBD

Main deck steam pipe renewal

OD 22mm X1000 mm - 6 pcs with both end flanges OD: 95mm

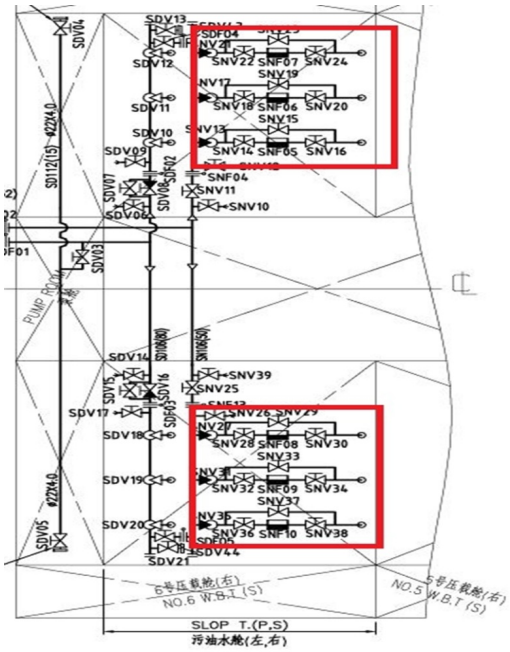
OD 32 mm X 100mm - 6 pcs With bith end flanges OD: 152mm/180mm

Pipe material: Steel SMLS or LSAW SCH. 80

Pipe Connection: Flange, Butt weld, sleeve couplings

Joints Flange rate:GB -10K, SLIP -ON

Gasket: SUS 304 Spiral Wound Graphite Gaskets



JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
6.SF.1.1	**Cost | Steel Tariff Rate**	2. Medium	Kilograms	Yard
	Description: ***all applicable surcharges to be specified.*			
6.SF.1.2	**Cost | Installing**	2. Medium	Lumpsum	Yard

6.SF.1 Steel Renewal | Air Line

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide service fees and units cost (kg) for the steel renewal works including all applicable surcharges.

Below mentioned general notes provide detailed instructions and considerations for steel renewal work during drydock.

General Notes

Edge Preparation and Grinding:
Defective steelwork edges are to be ground smooth and edge-prepared as required.

Bulkhead and Stiffener Renewal
Renewal of steel involves proper fairing, alignment, and witness "fit up" conditions.

Welding Practices
High tensile steel is to be pre-heated as per classification requirements.
Back gouging is to be seen by the Superintendent before final welding.
Full penetration welding with proper edge preparation is required.
Approved low hydrogen electrodes and welding sequences are to be used and submitted for approval.

Post-Weld Treatment
Steel fabrication and renewals are to be prepared before painting, including the removal of weld spatters, rounding sharp edges, dressing weld beads, and removing scars.

Hole and Slot Preparation
Edges of all holes, slots, etc., are to be ground smooth with no serration.

Grit Blasting and Coating
Fabricated steel sections are to be grit blasted to SA 2.5 ISO 8501-1, inspected, and coated ashore before delivery.
All steel used for deck and shell repairs is to be supplied on-board, blasted, and primed after fitting.

Scope

Air line from Aft to Midship crane / Supply pipes for Ballast Water Treatment System Lines Port and Stb. From Aft to Midship (Under catwalk)
To Renew wasted pipes

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
6.SF.1.1	**Cost | Steel Tariff Rate**	2. Medium	Kilograms	Yard
	Description: ***all applicable surcharges to be specified.*			
6.SF.1.2	**Cost | Installing**	2. Medium	Lumpsum	Yard
	Description: Air Supply Lines 50mm x 120 m , Flanges 50 pcs , Elbows 20 pcs , bolts and nuts 100 pcs 25mm x 35 meters , Flanges 30 pcs, Elbow 4 pcs , Bolt and nuts – 60 pcs Port and stb. Ballast supply pipes for BWTS – 50mm x 100 m, Flanges 100 pcs, elbows – 30 pcs.			

6.SF.1 Steel Renewal | PS Railings Manifold area

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide service fees and units cost (kg) for the steel renewal works including all applicable surcharges.

Below mentioned general notes provide detailed instructions and considerations for steel renewal work during drydock.

General Notes

Edge Preparation and Grinding:
Defective steelwork edges are to be ground smooth and edge-prepared as required.

Bulkhead and Stiffener Renewal
Renewal of steel involves proper fairing, alignment, and witness "fit up" conditions.

Welding Practices
High tensile steel is to be pre-heated as per classification requirements.
Back gouging is to be seen by the Superintendent before final welding.
Full penetration welding with proper edge preparation is required.
Approved low hydrogen electrodes and welding sequences are to be used and submitted for approval.

Post-Weld Treatment
Steel fabrication and renewals are to be prepared before painting, including the removal of weld spatters, rounding sharp edges, dressing weld beads, and removing scars.

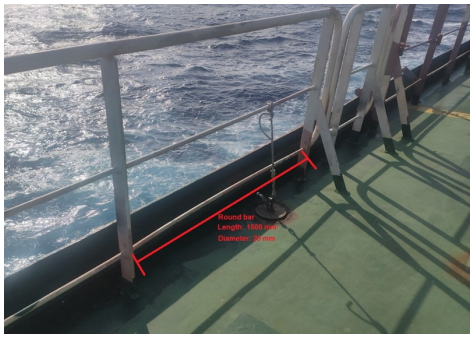
Hole and Slot Preparation
Edges of all holes, slots, etc., are to be ground smooth with no serration.

Grit Blasting and Coating
Fabricated steel sections are to be grit blasted to SA 2.5 ISO 8501-1, inspected, and coated ashore before delivery.
All steel used for deck and shell repairs is to be supplied on-board, blasted, and primed after fitting.

Scope

Please provide quotation for removal of damaged handrails, material supply, manufacture, and installation of new handrails as per original dimensions and material. After completion of installation, the surface of the new and burnt areas are to be mechanically prepared to ST 3.0 standard and coated with owner supplied paint.

Location: Port side midship near manifold area



JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
6.SF.1.1	**Cost | Steel Tariff Rate**	2. Medium	Kilograms	Yard
	Description: ***all applicable surcharges to be specified.*			
6.SF.1.2	**Cost | Installation**	2. Medium	Lumpsum	Yard
	Description: TO WELD A DIA 20MM X 1500MM ROUND BAR TO BOTH STANCHIONS			

6.SF.1 Steel Renewal | Expansion Tank Inspection Door

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

Please provide service fees and units cost (kg) for the steel renewal works including all applicable surcharges.

Below mentioned general notes provide detailed instructions and considerations for steel renewal work during drydock.

General Notes

Edge Preparation and Grinding:
Defective steelwork edges are to be ground smooth and edge-prepared as required.

Bulkhead and Stiffener Renewal
Renewal of steel involves proper fairing, alignment, and witness "fit up" conditions.

Welding Practices
High tensile steel is to be pre-heated as per classification requirements.
Back gouging is to be seen by the Superintendent before final welding.
Full penetration welding with proper edge preparation is required.
Approved low hydrogen electrodes and welding sequences are to be used and submitted for approval.

Post-Weld Treatment
Steel fabrication and renewals are to be prepared before painting, including the removal of weld spatters, rounding sharp edges, dressing weld beads, and removing scars.

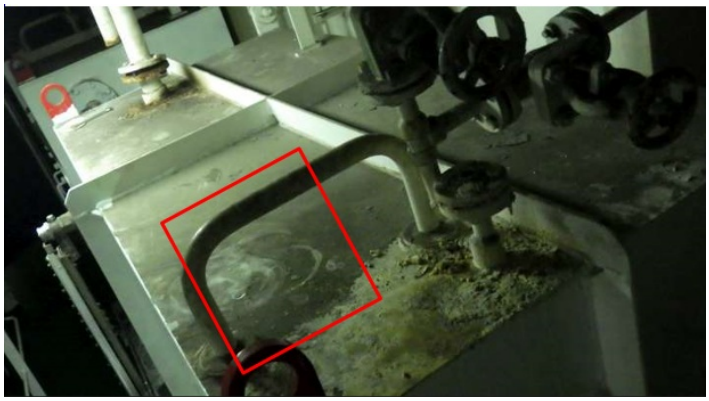
Hole and Slot Preparation
Edges of all holes, slots, etc., are to be ground smooth with no serration.

Grit Blasting and Coating
Fabricated steel sections are to be grit blasted to SA 2.5 ISO 8501-1, inspected, and coated ashore before delivery.
All steel used for deck and shell repairs is to be supplied on-board, blasted, and primed after fitting.

Scope

Please provide quotation for the installation of an inspection door on the HT and LT Expansion Tanks

Location: Engine Room Funnel Casing (B-Deck)



JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
6.SF.1.1	**Cost | Steel Tariff Rate**	2. Medium	Kilograms	Yard
	Description: ***all applicable surcharges to be specified.*			
6.SF.1.2	**Cost | Installation**	2. Medium	Lumpsum	Yard

Description:

TO WELD A DIA 20MM X 1500MM ROUND BAR TO BOTH STANCHIONS

7.BT.1 Ballast Tank Preparation & Equipment Mobilization

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

General

Preparation
1) Substrate preparation shall be carried out in accordance with ISO 8501–4–2006.
2) If steel repairs are to be carried out all rough edges, welding spatters, uneven bead, all scares left by the removal of fairing aids, etc. are to be removed and ground smooth and flush prior to commencement of the actual surface preparation work at no additional costs.
Spot Blasting
1) Spot Blasting shall be in accordance with ISO 8501–1–2007.
2) All coating edges of spot blasted areas are to be feathered edged either by blasting or power tooling.
Coating Application
1) Coating application may only be undertaken on the instructions of the superintendent
2) Full coating may only be undertaken during the hours of daylight
3) Coating application will be subject to humidity substrate temperature and inter coating period but in general terms coatings will not be applied when the substrate temperature is less than 3 degrees centigrade below the dew point temperature
4) Coatings will be applied using airless spray in accordance with the coating supervisor's instructions with
particular reference to spray gun fan and nozzle dimensions.
5) All coatings to be correctly stored and mixed prior to application. Thinners may not be added to coatings at any stage of the application
6) Typical areas where stripe coats should be applied include behind bars, plate edges, cut outs i.e. scallops, manholes etc. welds, areas of difficult access, ladders, behind face plates of web frames, etc. in accordance with coating supervisor's recommendations.
Inspection and Supervision
1) Prior to coating application, all blasted areas to be inspected and accepted by the superintendent and his designated supervisors.
2) The shipyard is to ensure the timely advice of blast inspection times.
3) Substrate inspections will only be taken during the daylight hours.
4) The owner does not accept responsibility or costs for any delays caused through rejection of blasted substrates or flash rusting as a consequence of incorrect edge preparation.
5) All paint spray applicators are to be provided with wet film thickness combs to ensure coating are being applied at the correct film thickness.
6) The superintendent retains the right to inspect coating thickness at any time during the application.
7) All paint and solvents as well as all the paint equipment should be properly and safely stored, out of direct sunlight so that the temperature of the materials will not exceed 35 C for prolonged periods of time. Strict control over paint consumption is to be maintained throughout the complete coating phase for each specified tanks after each stripe and full coat application and reported regularly to superintendent.

Scope

General Surface area + volume of each Ballast + FPT +APT

FPK – 947 m3 / 1P -1920m3 / 1S-2196m3 / 2P-1628m3 / 2S-1904m3 / 3P-1490m3 / 3S-1766m3 / 4P-1490m3 / 4S-1767m3 / 5P-1470m3 / 5S-1746m3 / 6P-1761 m3 / 6S-2095m3 / WB.ST.-132m3 / TCWT-276m3

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
7.BT.1.1	**Mobilization of Equipment**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for mobilization of washing, blasting, and coating equipment and demobilization after completion of works and final inspections. Quantity: 1.0 lump–sum Cost to include temporary ventilation and lighting during coating repairs and disconnection after completion of works and final inspections for 13 tanks			
7.BT.1.2	**Cost | Crane Service**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for crane service for coating repairs. Quantity: 1.0 – lump–sum			
7.BT.1.3	**Daily Gas Free Inspection**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for daily gas free inspections including submission of certificates for all tanks to be coated before and during washing, blasting, and coating. Quantity: 1.0 – lump–sum			
7.BT.1.4	**Cost | Tank Protection**	2. Medium	Tanks	Yard
	Description: Please provide quotation for protection of vessel equipment/ladders/pipes/anodes etc. inside ballast tanks. Quantity: 13 – tanks			
7.BT.1.5	**Cost | Manholes Open & Close**	2. Medium	Pieces	Yard
	Description: Please provide quotation for opening manholes and closing with new shipyard supplied gaskets. Quantity: 26 – pcs Cost to include quotation for assembling temporary safety cages around open manholes during coating repairs and disassembling after completion of works and final inspections. Quantity: 26 – pcs			
7.BT.1.6	**Tank Cleaning & Waste Disposal**	2. Medium	Lumpsum	Yard

Description:

1) Please provide quotation for collection and disposal of empty paint drums as per local and international environmental regulations

Quantity: 1.0 – lump–sum

2) Please provide quotation for collection and disposal of washing/blasting water as per local and international environmental regulations

Quantity: 13.0 – tanks

7.CT.1 Cargo Tanks Coating & Treatment

Vessel	SEAWAYS TRITON	**Location**	
Type/serial no		**Component**	
Start date	12.03.2025	**End date**	28.02.2026
Status	Not started	**Priority**	2. Medium
Responsible	Jatinder Singh	**Maker**	

DESCRIPTION

GENERAL

Preparation

1) Substrate preparation shall be carried out in accordance with ISO 8501–4–2006.

2) If steel repairs are to be carried out all rough edges, welding spatters, uneven bead, all scares left by the removal of fairing aids, etc. are to be removed and ground smooth and flush prior to commencement of the actual surface preparation work at no additional costs.

Spot Blasting

1) Spot Blasting shall be in accordance with ISO 8501–1–2007.

2) All coating edges of spot blasted areas are to be feathered edged either by blasting or power tooling.

Coating Application

1) Coating application may only be undertaken on the instructions of the superintendent

2) Full coating may only be undertaken during the hours of daylight

3) Coating application will be subject to humidity substrate temperature and inter coating period but in general terms coatings will not be applied when the substrate temperature is less than 3 degrees centigrade below the dew point temperature

4) Coatings will be applied using airless spray in accordance with the coating supervisor's instructions with particular reference to spray gun fan and nozzle dimensions.

5) All coatings to be correctly stored and mixed prior to application. Thinners may not be added to coatings at any stage of the application

6) Typical areas where stripe coats should be applied include behind bars, plate edges, cut outs i.e. scallops, manholes etc. welds, areas of difficult access, ladders, behind face plates of web frames, etc. in accordance with coating supervisor's recommendations.

Inspection and Supervision

1) Prior to coating application, all blasted areas to be inspected and accepted by the superintendent and his designated supervisors.

2) The shipyard is to ensure the timely advice of blast inspection times.

3) Substrate inspections will only be taken during the daylight hours.

4) The owner does not accept responsibility or costs for any delays caused through rejection of blasted substrates or flash rusting as a consequence of incorrect edge preparation.

5) All paint spray applicators are to be provided with wet film thickness combs to ensure coating are being applied at the correct film thickness.

6) The superintendent retains the right to inspect coating thickness at any time during the application.

7) All paint and solvents as well as all the paint equipment should be properly and safely stored, out of direct sunlight so that the temperature of the materials will not exceed 35 C for prolonged periods of time. Strict control over paint consumption is to be maintained throughout the complete coating phase for each specified tanks after each stripe and full coat application and reported regularly to superintendent.

Ballasting and coating areas (Estimated as per sister vessel)

All paint manufacturers' technical data sheets are to be strictly followed

during washing, blasting, and coating applications without any exception. Final blasting areas are to be decided after detailed inspection by the coating supervisor. Please prepare the quotation considering both full and spot blasting areas.

TO BE INCLUDED:
- Cleaning after
- Cleaning before
- Corrosion protection
- Gas free certificate
- Lighting
- Paintwork
- Staging

MATERIALS TO BE SUPPLIED BY:
- Owner's supply

THE WORK TO BE SURVEYED BY:
- Makers representative
- Owners representative

JOB ITEMS

SFI*	Item Name	Priority	Type	Cost type
7.CT.1.2	**Daily Gas Free Inspections**	2. Medium	Lumpsum	Yard
	Description: Please provide quotation for daily gas free inspections including submission of certificates for all tanks to be coated before and during washing, blasting, and coating.			
7.CT.1.3	**Protection of Tanks**	2. Medium	Tanks	Yard
	Description: 1) Please provide quotation for protection of ladders, pipes, etc. inside cargo tanks. 2) Please provide quotation for protection of heating coils inside cargo tanks.			
7.CT.1.4	**Temporary Lighting**	2. Medium	Tanks	Yard
	Description: Please provide quotation for connection and usage of temporary lighting during coating repairs and disconnection after completion of works and final inspections.			
7.CT.1.5	**Temporary Ventilation**	2. Medium	Tanks	Yard
	Description: Please provide quotation for connection and usage of temporary ventilation during coating repairs and disconnection after completion of works and final inspections.			